
End-to-End Deep Reinforcement Learning Platform for Competitive Physical Dynamic Systems

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Abstract

This work introduces a new benchmark framework for evaluating reinforcement learning (RL) agents in competitive multi-agent environments in both simulated and physical systems. The framework allows agents to be trained locally in simulation and subsequently tested on standardized hardware, enabling analysis of performance transfer from virtual to real-world conditions. To facilitate consistent and reproducible evaluations, the framework provides standardized specifications and a centralized leaderboard that competitively compares agent performance across domains. This platform is designed to advance the development of competitive agents capable of robust, adaptive behavior and effective sim-to-real generalization, serving as a standardized resource for the broader RL community.

1 Introduction

Foosball is a fast-paced, strategy-driven table game that requires sub-second decision-making, fine motor control, and dynamic tactics. Reinforcement Learning (RL) techniques are widely adopted for automating dynamic control tasks because they can discover near-optimal policies across diverse environments Margolis et al. [2022], Tang et al. [2024], Kober et al. [2013]. To train the fast, physical skills involved in foosball effectively, an RL agent must learn within a simulator that faithfully reproduces the sport’s real-world, high-speed dynamics, ensuring the resulting policy transfers to

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an actual table D’Ambrosio et al. [2023], Kaufmann et al. [2023]. Furthermore, choosing an RL algorithm well suited to these rapid, interactive settings is essential for maximizing decision-making performance Kurach et al. [2020], Tammewar et al. [2023].

In this work, we aim to explore the domain of sim-to-real transfer in foosball. We begin by constructing a MuJoCo-based physics simulation whose kinematic and contact constraints closely mirror the actual foosball table used in our lab. We created 3D models of the foosball table components with real-world dimensions to accurately reconstruct them in the simulation environment. Subsequently, constraint modeling was performed by setting appropriate parameters to achieve physically realistic interactions during foosball playing. After training inside this environment, we present preliminary benchmarks of three reinforcement-learning algorithms: Twin Delayed Deep Deterministic Policy Gradient (TD3) Fujimoto et al. [2018], Proximal Policy Optimization (PPO) Schulman et al. [2017], and Soft Actor-Critic (SAC) Haarnoja et al. [2019]. We measure how reliably each can score when domain randomization is applied and training resources are held constant.

Afterwards, we move towards the larger goal of constructing a physical mechanism that can be equipped to a standardized foosball table and enable physical manipulation of the foosball rods by AI agents. Our pipeline is set up to feed raw images directly into a Deep RL Agent, eliminating intermediate steps such as ball localization or rod-angle estimation. We propose this as a physical foosball platform that can be used to compare RL agents trained in separate simulated environments.

We introduce an end-to-end pipeline that enables researchers to train agents in simulation and evaluate them on standardized physical hardware. The system includes a high-fidelity simulation model, a physical foosball table with controllable actuation and automated ball return, and a global leaderboard that spans both simulated and real-world performance. Specifics on the digital and physical models are provided for ease of replication. We also benchmark several popular RL algorithms to assess their ability to in simulation. We conclude with a discussion of the benchmark’s potential for advancing sim-to-real research in competitive multi-agent settings.

2 Related Work

Research on autonomous foosball spans both simulated settings and physical tables. The studies summarized below highlight key progress and open challenges.

2.1 Game-State Estimation

Several projects extract the match state from camera images of the foosball surface Gutierrez-Franco et al. [2013], Hernández et al. [2019], Weigel and Nebel [2003]. Janssen *et al.* showed that a combination of regions of interest (ROI) and Kalman filtering can reliably track the ball in a known arena at speeds up to 10 m/s—even with rebounds and sudden direction changes Janssen et al. [2009]. More recently, CNN-based approaches have inferred rod positions and ball locations directly from images, underscoring deep networks’ capacity to recover a foosball game state without hand-crafted features Hagens et al. [2024], Horst et al. [2024].

2.2 Action Selection

Early autonomous players relied on hand-tuned decision trees for rod manipulation Weigel and Nebel [2003], Weigel [2005]. Imitation-learning pipelines soon followed, capturing advanced moves such as the “lock” and “slide-kick” techniques Zhang and Nebel [2007]. With the rise of Deep Reinforcement Learning (DRL), researchers framed scoring with a single rod as a control problem akin to a manufacturing cell De Blasi et al. [2020]. Subsequent studies extended DRL to individual skills such as dribbling and passing Croenen et al., and to multi-agent scenarios in which strikers and goalkeepers learn competing offensive and defensive policies from scratch Gashi et al. [2023], Pinto et al. [2017].

2.3 Sim-to-Real Transfer

Many works build physical foosball robots and then deploy the policies trained in simulation on the real hardware Moos et al. [2024], De Blasi et al. [2020], Weigel [2005], Rohrer et al. [2021]. This line of research sits within the broader Sim-to-Real literature for robotic manipulation Peng et al. [2017b,a],

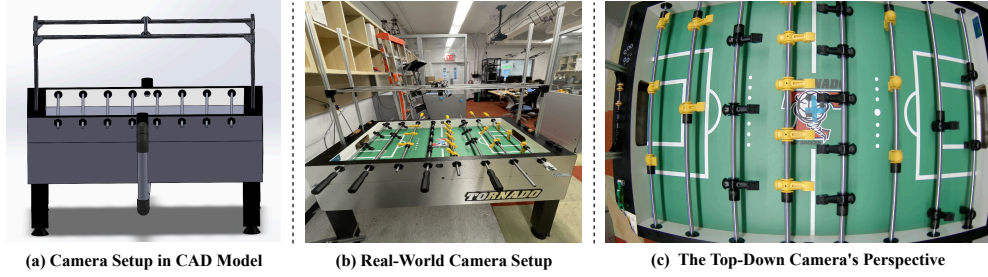


Figure 1: Camera Configuration for The AI-Controlled Foosball System

Tobin et al. [2017]. Reports consistently note a marked performance drop when control policies leave the simulator Moos et al. [2024], Rohrer et al. [2021]. Domain randomization—systematically varying simulation parameters—has been proposed as a hedge against overfitting to the virtual environment De Blasi et al. [2020], Peng et al. [2017a].

3 Mechanical Setup

3.1 Foosball Table

The experimental environment utilizes the Tornado T-3000 Commercial-Style Foosball Table, which is the official model of the International Table Soccer Federation (ITSF). This table was selected for its durability, precision engineering, and standardized dimensions, which ensure consistent and reproducible testing conditions. The selection of this official table model allows us to easily create 3D models that are used to build a physics simulation environment.

In contrast to many existing automated foosball systems, our design does not incorporate sensors on the field to track the ball or player rods. Instead, it relies solely on RGB image streams. This setup enables us to evaluate whether AI agents can learn to perceive and play foosball using only visual input.

3.2 Camera Setup

ELP 48MP USB cameras are used for vision in our setup. The camera has the following specifications that make it the ideal choice for our tasks:

- Resolution: 48MP
- Photo Sensor Technology: CMOS (Color Interpretation)
- FPS: 30 (Human eye 30 - 60)
- Global Shutter
- Focal Length: 3.6-10 mm

As shown in Figure 1 (a) and (b), the camera is mounted on a frame positioned 30 inches above the top surface of the foosball table, with its lens oriented downward to capture a full top-down view. The stand frames are constructed using 20 mm × 20 mm T-slotted aluminum extrusion rails, reinforced with aluminum corner bracket joint plates and L-brackets for stability. The camera itself is attached to an adjustable rail running along the center of the frame, allowing for fine-tuned positioning. This setup remains stable during the jarring motion of the foosball rods. Future experiments will be conducted to assess the reliability and robustness of this setup under more intensive playing conditions.

The rationale for selecting cameras with these specifications is that it closely replicates the visual capabilities of the human eye. This enables a realistic and effective environment for evaluating the performance of AI agents under human-like visual conditions. In the current setup, a single overhead ELP camera provides a full top-down view of the entire foosball table. If this current setup proves insufficient, we plan to add two more cameras to this configuration, allowing AI agents to have multiple perspectives. The camera view of the current configuration is shown in Figure 1 (c).

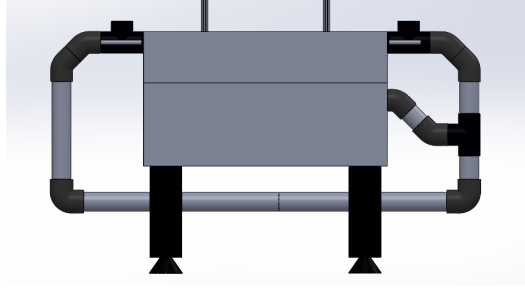


Figure 2: 3D CAD Design of The Ball Reset Mechanism

3.3 Ball Reset Mechanism

The current Ball Reset Mechanism prototype enables automatic ball return from one side of the foosball table. The proposed configuration utilizes 1.5-inch inner diameter PVC pipes, 3D-printed components, and a shop vacuum to create airflow that transports the ball. When a goal is scored, gravity guides the ball into a separate opening beneath the playing field. A 3D-printed funnel directs the ball into the PVC piping, where it is then propelled back onto the field by the vacuum-driven airflow.

In future iterations, we plan to expand this prototype to support a two-sided ball reset mechanism. This will more accurately replicate a real game setting, as ball return positions are offset on each side on the table to prevent dead balls and enable proper serving. To determine which side the next ball should be returned to, we intend to integrate sensors at each goal to track scoring events. This expanded design will require increased airflow capacity to maintain reliable ball transport. A proposed CAD model of the ball reset system is shown in Figure 2.

3.4 Mechanical Setup: Actuators and Control System

The actuator developed for the foosball table uses Teknic ClearPath-SC integrated servo motors (Model SCSKP), featuring an all-in-one architecture that combines a brushless rare-earth servo motor, high-resolution optical encoder, embedded servo drive, and trajectory generation unit within a compact NEMA 23 frame. These servo motors are employed to run actuators both in the rotational and translational axes of the system.

For translational motion, a belt-driven linear stage was used run by Teknic servo motor. This chosen mechanism provides high efficiency, very low backlash, and gives ability to deliver smooth linear motion.

3.4.1 Motor Selection

The actuators' dynamic and static requirements were chosen based on actual player performance, derived from gameplay analyses. For the translational motion, the primary challenge involved overcoming the combined effects of inertia and friction in the belt-driven transmission system. Experimental observations performed on the actual foosball table indicated a maximum acceleration of approximately 2.5 m/s^2 during peak gameplay is needed. The derived acceleration, coupled with the system mass and frictional forces, gave torque requirement of 0.45 Nm . The SCSK motor which offered continuous torque capacity of 0.75 Nm was selected, which provided a safety margin of 1.5.

Similarly for the rotational motion, rpm and torque demands were calculated based on gameplay analyses. These calculations showed a maximum torque requirement of around 0.25 Nm . The same servo motor was chosen for the rotational motion as well, simplifying system integration through hardware standardization across both motion types.

3.4.2 System Architecture

The control architecture developed for foosball table actuation supports up to four ClearPath-SC motors per SC4-Hub, with plan to reach 16 axes through using multiple hub configurations. Communication between the motors and the control system is established via USB 2.0, leveraging Teknic's

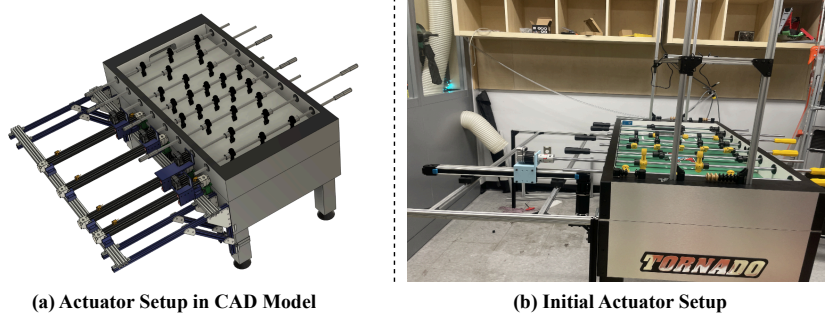


Figure 3: Actuator Configuration for Control System of the AI Multi Agents

sFoundation C++ library for real-time, multi-threaded control. Each motor is interfaced through a dedicated I/O connector for digital signaling, and diagnostic access is provided through a micro-USB port compatible with Teknic’s ClearView software, which facilitates motor tuning and performance monitoring.

3.4.3 Actuator Placement

Motor placement was carefully optimized to minimize structural overhang and reduce the mechanical loading on the frame, Figure 3 (a). The full physical assembly is being build. The initial assembly can be seen in Figure 3 (b). The linear actuator motor is mounted on the stationary base to limit moving mass, thereby improving system stability and reducing vibrational effects. Cable management was also considered, ensuring clear access to all ports and power interfaces while reducing the risk of mechanical interference during operation or any cable fatigue.

4 Digital Model of Table: MuJoCo

Foosball play involves rapid interactions between the ball and players. Given the features of fast movements of the ball and accurate contact state estimation during the play, MuJoCo (Multi-Joint dynamics with Contact) physics simulator is utilized to conduct the simulation training in this study.

4.1 MuJoCo Simulator

MuJoCo is an open-source physics engine designed to facilitate advanced scientific research and development in domains such as robotics, biomechanics, graphics, and animation. The simulator provides various constraints and configurable options for contact and friction dynamics, enabling accurate contact estimation. Its capacity for both precise and rapid simulations makes it particularly well-suited for applications that require accuracy and computational efficiency, such as foosball playing Todorov et al. [2012].

4.2 Object Rendering using SDF

The 3D models of the official foosball table components were created to construct the MuJoCo simulation environment using Signed Distance Field (SDF). SDF is a spatial domain representation where each sampled point in a grid encodes the shortest distance to the surface of an object’s geometry. SDF is widely used for rendering complex 3D shapes in simulations that require accurate boundary representation of rigid bodies for precise collision detection Jones et al. [2006]. The surface boundary of the object is defined by the sign of the value indicating whether the point is inside or outside the shape, as illustrated in Figure 4 (a). The mathematical expression of SDF is:

$$SDF(p) = \text{sign}(p) \cdot \text{dist}(p)$$

Where p represents a query point on a grid. The $\text{sign}(p)$ is either +1 or -1, indicating whether the point lies outside or inside the object’s surface geometry, respectively. The $\text{dist}(p)$ denotes the minimum

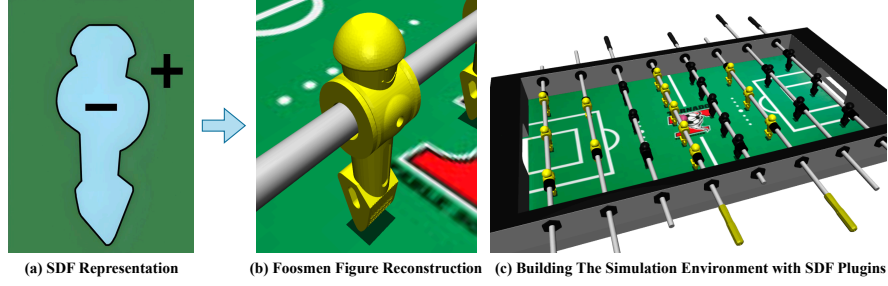


Figure 4: Foosball Gameplay Simulation Environment with Signed Distance Field Representation

Euclidean distance from the point p to the object’s boundary. Rendering in a simulation environment is accomplished by computing the signed distance field and its gradient at specified query points.

MuJoCo offers the SDF plugin functionalities that allow for importing customized 3D models with accurate geometry representations. The function `sdf_aabb` in the `sdf` library computes the axis-aligned bounding box in local coordinates. This volume is uniformly voxelized before calling to the marching cubes algorithm. The table and foosmen are rendered using the SDF plugins, enabling precise collision detection between the ball, the foosmen, and the table. Figure 4 (b) and (c) show the appearance of the foosmen figure and the whole MuJoCo simulation environment through the SDF rendering of objects, respectively.

4.3 Constraint Model

MuJoCo supports several types of constraint models, governing physical interactions between objects. The constraint is activated and numerically enforced when a violation occurs, such as exceeding joint limits or penetration in collisions, to simulate real-world physics.

4.3.1 Solver Parameters

There are three constraint solver parameters under user’s control, d , b , k , which determine the behaviors of the constraint models in the simulation. These parameters can be set indirectly through the attributes available in MJCF (MuJoCo XML Format) elements. A new constraint formulation is used in MuJoCo to define the constraint. A single scalar expression of constraint is:

$$a_1 + d \cdot (bv + kr) = (1 - d) \cdot a_0$$

Where a_0 represents the acceleration in the absence of constraint force, and a_1 the acceleration governed by a constraint model. v refers to the velocity, r the position deviation or residual. b and k denote the damping and stiffness of the virtual spring-damper systems, respectively. Here, $d \in (0, 1)$ is defined as the constraint impedance which interpolates between free acceleration and constraint-induced acceleration.

The function of impedance $d(r)$ is determined by the parameter vector in `solimp` attribute: `solimp = (d0, dwidth, width)`. In this function, r represents the range of `width` that determines the sigmoidal function $d(r)$, where $d(0) = d_0$ and $d(width) = d_{width}$. Outside of the range, the impedance has a constant value $d(0)$. Given the collisions between two rigid bodies in a foosball play, we generate a strong constraint model by setting impedance d_0 and d_{width} as 0.99 and r as 0.0001.

The damping b and stiffness k are set using the `solref` attribute: `solref = (timeconst, dampratio)`, parameterizing the model based on the time constant and a mass-spring-damper system. The damping and stiffness are defined as:

$$b = 2 / (d_{width} \cdot timeconst)$$

$$k = d(r) / (d_{width}^2 \cdot timeconst^2)$$

To ensure rapid recovery from constraint violations, we set the time constant to 0.004 and damping ratio to 1.0 (critical damping). We are further exploring the parameters in the attribute to achieve a

balance between the stiffness and damping for the realistic movement of the ball. And by combining the above formulas with the constraint expression, the following result is derived when the impedance d is constant:

$$r = a_0 \cdot (1 - d) \cdot \text{timeconst}^2 \cdot \text{dampratio}^2$$

As setting a small time constant, only little penetration depth is allowed when collisions between two rigid bodies take place.

4.3.2 Contact Modeling

In MuJoCo, contact is the most elaborate type of constraints within the model, making contact modeling particularly challenging. Point contacts are utilized in MuJoCo, which are defined by a point between two geometries and a spatial frame at the center of the point. The dimensionality of the contact force is defined using the *condim* attribute. We set *condim* = 3, which represents a standard frictional contact and generates normal force and a tangential friction force opposing slip.

4.3.3 Friction Loss

Setting friction loss also plays a crucial role in foosball simulation, as it enables the real motion of the ball on the floor. Friction loss, also referred to as dry friction, static friction, or load-independent friction, is different from damping or viscosity in that a velocity-dependent force is applied in the opposite direction of the ball’s motion. This constant force enables the ball to move realistically on the floor, allowing it to stop smoothly. We are currently exploring the appropriate parameters for the ball friction loss.

5 RL Platform and Agents

We evaluated three widely used continuous-control reinforcement-learning algorithms under a unified experimental framework. All agents were implemented with the Stable Baselines 3 library Raffin et al. [2021], ensuring consistent game state. Each policy employed the same multilayer-perceptron (MLP) backbone: seven fully connected layers of 3 000 hidden units. Adam optimizers with identical learning-rate schedules were applied across algorithms, and gradient clipping was enabled to stabilize updates. Below we provide a quick review of the features of each method.

5.1 Proximal Policy Optimization (PPO) Schulman et al. [2017]

Proximal Policy Optimization is an on-policy, actor–critic algorithm designed to balance between sample efficiency and implementation simplicity. During each update, PPO maximizes the clipped objective:

$$L_{\text{clip}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon) A_t) \right], \quad r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)},$$

which limits the magnitude of policy updates and preventing destructive oscillations.

5.2 Soft Actor–Critic (SAC) Haarnoja et al. [2019]

Soft Actor–Critic is an off-policy algorithm that augments the conventional actor–critic framework with an entropy-regularization term, encouraging the policy to remain stochastic and explore the action space:

$$J_\pi = \mathbb{E}_{s \sim \mathcal{D}} \left[\mathbb{E}_{a \sim \pi(\cdot | s)} [Q(s, a) - \alpha \log \pi(a | s)] \right].$$

Two independent critics are trained to minimize a squared-Bellman loss, while a temperature parameter α is tuned automatically to match a target entropy.

5.3 Twin Delayed Deep Deterministic Policy Gradient (TD3) Fujimoto et al. [2018]

TD3 extends Deep Deterministic Policy Gradient by mitigating overestimation bias through three modifications: (i) maintaining twin critic networks and taking the minimum of their target values,

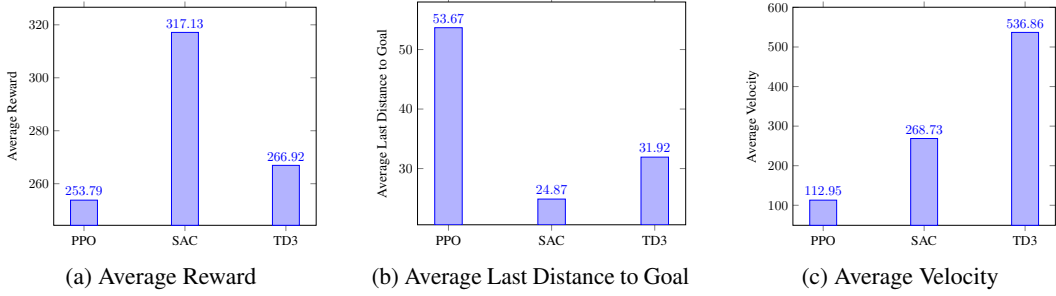


Figure 5: Evaluation metrics computed over 10^3 deterministic episodes.

(ii) delaying actor updates with respect to critic updates, and (iii) adding clipped Gaussian noise to the target action:

$$y = r + \gamma \min_{i \in \{1,2\}} Q_{\phi'_i}(s', \pi_{\theta'}(s') + \varepsilon), \quad \varepsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c).$$

In our study, critic updates outnumbered actor updates by a factor of two, and target-policy smoothing used $\sigma = 0.2$, $c = 0.5$.

6 Training Protocol

All three algorithms were trained in the foosball environment introduced in Section 3. Each run comprised 15 epochs, where one epoch equaled 10^5 simulation time steps. A replay buffer of capacity 10^6 was pre-allocated and cleared only between independent seeds. To track out-of-sample performance, we performed deterministic evaluations every 3 000 steps, averaging returns over ten held-out episodes with exploration disabled. Hyperparameters not listed above—such as learning rates (3×10^{-4} for actors and critics), discount factor ($\gamma = 0.99$), and gradient norms—were held constant across agents. All experiments executed on a single NVIDIA RTX 4090 GPU.

7 Preliminary Results

Each trained controller was subjected to a test phase comprising 10^3 independent episodes in the same simulation environment, with exploration noise disabled so that actions were chosen deterministically. For every episode we logged the full trajectory of states, actions, and rewards, and afterward computed the following diagnostics:

- **Score Rate:** the proportion of episodes in which the ball ultimately crossed the opponent’s goal line.
- **Average Distance to Goal:** the mean Euclidean distance between the ball and the goal center, aggregated over *all* time steps in all evaluation episodes.
- **Average Velocity:** the mean speed of the ball, again averaged over every time step across the evaluation set.
- **Average Reward:** the mean cumulative reward obtained per episode under the deterministic policy.
- **Average Last Distance to Goal:** the ball-to-goal distance measured at the *final* time step of each episode, then averaged over episodes.

Overall performance. Under the chosen training horizon and domain-randomization settings, **SAC emerged as the top-performing algorithm**. As depicted in Fig. ??, SAC achieved a 42.9% Score Rate, more than doubling the values recorded for PPO and TD3. Correspondingly, its Average Reward was noticeably higher, reflecting the reward bonus associated with a successful goal.

Proximity versus conversion. Although SAC converted the largest share of episodes into goals, Fig. 5b reveals that TD3 ended its episodes almost as close to the target. The Average Last Distance

to Goal of TD3 was 31.92 versus 24.87 for SAC. This indicates that TD3 learned to advance the ball into shooting range but lacked a reliable finishing tactic. Given the relatively short 15-epoch budget, it is likely that additional training would allow TD3 to narrow, or even close, the scoring gap.

Ball-handling dynamics. Interestingly, the agents discovered distinct ball-speed regimes despite the absence of an explicit velocity term in the objective. TD3 propelled the ball at roughly twice the speed attained by SAC, which itself moved the ball about twice as fast as PPO. These differences are reflected in the *Average Velocity* metric and suggest that each algorithm gravitated toward a unique trade-off between control precision and shot power.

In summary, the evaluation confirms that SAC currently offers the most reliable end-to-end policy under our training constraints, while TD3 shows promise in ball advancement that could translate into higher scoring with extended training. PPO, although stable, lags behind both in conversion efficiency and ball-movement aggressiveness.

8 Benchmark Suitability

Foosball presents a unique and compelling testbed for reinforcement learning research, particularly in the context of sim-to-real generalization and multi-agent interaction. The game combines several challenging attributes: high-speed, dynamic control; mechanical interactions; and adversarial gameplay requiring adaptive strategic behavior. These elements provide a balanced mixture of low-level motor control, high-level policy learning, and opponent modeling.

Unlike some RL domains that operate in high-dimensional, unpredictable environments, foosball provides a controlled setting with complex, yet well-defined dynamics. The game’s physical structure and rule set are easy to learn and replicate both in simulation and in laboratory setups. At the same time, the need for precise timing, coordination across multiple rods, and rapid decision-making under adversarial pressure makes it a rich test case for learning across various RL frameworks as can be seen in the preceding section. These characteristics allow for meaningful evaluation of an agent’s ability to generalize across domains without being overwhelmed by noise or large mechanical movements.

To support standardized evaluation and enable reproducible research, the benchmark includes a centralized leaderboard that records agent performance in both simulation and on physical hardware. The leaderboard ranks agents based on competitive metrics such as win rate, goal differential, and time-to-score, with optional evaluation against a growing pool of previously submitted opponents. This setup allows researchers to not only assess absolute performance but also understand how well their agents generalize across different play styles.

To enable continuous physical testing of new agents, the foosball hardware integrates an automated ball return and injection system. This allows for uninterrupted gameplay without human intervention, making it feasible to evaluate large numbers of agents reliably. Additionally, the system enforces predefined limits on rod translation and rotation speeds, ensuring that all agents operate within the physical constraints of the real table, minimizing the probability of off table shots. These measures guarantee consistent, repeatable testing conditions and minimize the risk of hardware damage, allowing the physical environment to serve as a robust and scalable evaluation platform.

9 Conclusion

This paper introduces a complete benchmark framework for evaluating RL in a competitive foosball setting, bridging both simulation and physical environments. The system combines a high-fidelity simulation model with a physical foosball platform equipped with vision and mechanical resets. By supporting reproducible deployment pipelines and standardized evaluation protocols, the benchmark allows agents trained in simulation to be tested on real-world hardware with minimal human intervention. A leaderboard infrastructure tracks agent performance across domains, enabling fair comparisons and long-term progress tracking. Foosball’s structured yet dynamic gameplay presents a well-suited environment for studying sim-to-real generalization, competitive RL strategies, and control robustness.

References

- Jonathan Croenen, Jens Bürger, and Wannes Meert. Reinforcement learning of action sequences in table football. URL <https://api.semanticscholar.org/CorpusID:276604268>.
- Stefano De Blasi, Sebastian Klöser, Arne Müller, Robin Reuben, Fabian Sturm, and Timo Zerrer. Kicker: An industrial drive and control foosball system automated with deep reinforcement learning, 04 2020.
- David D’Ambrosio, Navdeep Jaitly, Vikas Sindhwani, Ken Oslund, Peng Xu, Nevena Lazic, Anish Shankar, Tianli Ding, Jonathan Abelian, Erwin Coumans, Gus Kouretas, Thinh Nguyen, Justin Boyd, Atıl İscen, Reza Mahjourian, Vincent Vanhoucke, Alex Bewley, Yuheng Kuang, Michael Ahn, Deepali Jain, Satoshi Kataoka, Omar Cortes, Pierre Sermanet, Corey Lynch, Pannag Sanketi, Krzysztof Choromanski, Wenbo Gao, Juhana Kangaspunta, Krista Reymann, Grace Vesom, Sherry Moore, Avi Singh, Saminda Abeyruwan, and Laura Graesser. Robotic table tennis: A case study into a high speed learning system. In *Robotics: Science and Systems XIX*, RSS2023. Robotics: Science and Systems Foundation, July 2023. doi: 10.15607/rss.2023.xix.006. URL <http://dx.doi.org/10.15607/RSS.2023.XIX.006>.
- Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods, 2018. URL <https://arxiv.org/abs/1802.09477>.
- Adriatik Gashi, Elke Hergenröther, and Gunter Grieser. Efficient training of foosball agents using multi-agent competition. In Kohei Arai, editor, *Intelligent Computing*, pages 472–492, Cham, 2023. Springer Nature Switzerland. ISBN 978-3-031-37717-4.
- Juan Gutierrez-Franco, John Inlow, Jesse Graham, and Larry Huang. Automated foosball table year 1: Final project report. 12 2013.
- Tuomas Haarnoja, Aurick Zhou, Kristian Hartikainen, George Tucker, Sehoon Ha, Jie Tan, Vikash Kumar, Henry Zhu, Abhishek Gupta, Pieter Abbeel, and Sergey Levine. Soft actor-critic algorithms and applications, 2019. URL <https://arxiv.org/abs/1812.05905>.
- David Hagens, Jan M. Knaup, Elke Hergenröther, and Andreas Weinmann. Cnn-based game state detection for a foosball table, 2024. URL <https://arxiv.org/abs/2404.05357>.
- Nicolás Hernández, Javier Rebolledo, Javier Torres, Gonzalo Carvajal, and Francisco Vargas. Design of an experimental platform for the automation of the goalkeeper of a foosball table. In *2019 IEEE CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON)*, pages 1–7, 2019. doi: 10.1109/CHILECON47746.2019.8988113.
- Ronny Horst, David Hagens, Elke Hergenröther, and Andreas Weinmann. Real time state detection of a foosball game using cnn-based computer vision. In Kohei Arai, editor, *Intelligent Computing*, pages 343–354, Cham, 2024. Springer Nature Switzerland. ISBN 978-3-031-62281-6.
- Rob Janssen, Jeroen Best, and M.J.G. Molengraft. Real-time ball tracking in a semi-automated foosball table. pages 128–139, 06 2009. ISBN 978-3-642-11875-3. doi: 10.1007/978-3-642-11876-0_12.
- M.W. Jones, J.A. Baerentzen, and M. Sramek. 3d distance fields: a survey of techniques and applications. *IEEE Transactions on Visualization and Computer Graphics*, 12(4):581–599, 2006. doi: 10.1109/TVCG.2006.56.
- Elia Kaufmann, Leonard Bauersfeld, Antonio Loquercio, Matthias Müller, Vladlen Koltun, and Davide Scaramuzza. Champion-level drone racing using deep reinforcement learning. *Nature*, 620(7976):982–987, 2023.
- Jens Kober, J. Andrew Bagnell, and Jan Peters. Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research*, 32(11):1238–1274, 2013. doi: 10.1177/0278364913495721. URL <https://doi.org/10.1177/0278364913495721>.
- Karol Kurach, Anton Raichuk, Piotr Stańczyk, Michał Zajac, Olivier Bachem, Lasse Espeholt, Carlos Riquelme, Damien Vincent, Marcin Michalski, Olivier Bousquet, and Sylvain Gelly. Google research football: A novel reinforcement learning environment, 2020. URL <https://arxiv.org/abs/1907.11180>.
- Gabriel B Margolis, Ge Yang, Kartik Paigwar, Tao Chen, and Pulkit Agrawal. Rapid locomotion via reinforcement learning, 2022. URL <https://arxiv.org/abs/2205.02824>.
- Janosch Moos, Cedric Derstroff, Niklas Schröder, and Debora Clever. Learning to play foosball: System and baselines. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, page 4326–4332. IEEE, May 2024. doi: 10.1109/icra57147.2024.10611321. URL <http://dx.doi.org/10.1109/ICRA57147.2024.10611321>.

- Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and P. Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1–8, 2017a. URL <https://api.semanticscholar.org/CorpusID:3707478>.
- Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. *CoRR*, abs/1710.06537, 2017b. URL <http://arxiv.org/abs/1710.06537>.
- Lerrel Pinto, James Davidson, Rahul Sukthankar, and Abhinav Gupta. Robust adversarial reinforcement learning. *CoRR*, abs/1703.02702, 2017. URL <http://arxiv.org/abs/1703.02702>.
- Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. Stable-baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22 (268):1–8, 2021. URL <http://jmlr.org/papers/v22/20-1364.html>.
- Tobias Rohrer, Ludwig Samuel, Adriatik Gashi, Gunter Grieser, and Elke Hergenröther. Foosball table goalkeeper automation using reinforcement learning. In *Lernen, Wissen, Daten, Analysen*, 2021. URL <https://api.semanticscholar.org/CorpusID:240290331>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Akshaj Tammewar, Nikita Chaudhari, Bunny Saini, Divya Venkatesh, Ganpathiraju Dharahas, Deepali Vora, Shruti Patil, Ketan Kotecha, and Sultan Alfarhood. Improving the performance of autonomous driving through deep reinforcement learning. *Sustainability*, 15(18), 2023. ISSN 2071-1050. doi: 10.3390/su151813799. URL <https://www.mdpi.com/2071-1050/15/18/13799>.
- Chen Tang, Ben Abbatematteo, Jiaheng Hu, Rohan Chandra, Roberto Martín-Martín, and Peter Stone. Deep reinforcement learning for robotics: A survey of real-world successes, 2024. URL <https://arxiv.org/abs/2408.03539>.
- Joshua Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and P. Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. URL <https://api.semanticscholar.org/CorpusID:2413610>.
- Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. doi: 10.1109/IROS.2012.6386109.
- T. Weigel. Kiro - a table soccer robot ready for the market. In *Proceedings of the 2005 IEEE International Conference on Robotics and Automation*, pages 4266–4271, 2005. doi: 10.1109/ROBOT.2005.1570776.
- Thilo Weigel and Bernhard Nebel. Kiro – an autonomous table soccer player. In Gal A. Kaminka, Pedro U. Lima, and Raúl Rojas, editors, *RoboCup 2002: Robot Soccer World Cup VI*, pages 384–392, Berlin, Heidelberg, 2003. Springer Berlin Heidelberg. ISBN 978-3-540-45135-8.
- Dapeng Zhang and Bernhard Nebel. Learning a table soccer robot a new action sequence by observing and imitating. pages 61–, 01 2007.

A Technical Appendices and Supplementary Material

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